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Industrial Intelligence Systems: A Structured Review of Key Technologies, System Dependencies, and Engineering Trustworthiness

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Abstract: Industrial intelligence relies on the integration of industrial software, interconnected production systems, digital twins, and artificial intelligence (AI) to support data-driven monitoring, predictive analysis, and operational decision-making. Despite advances in these technologies, differences in technological maturity, system interoperability, and engineering trustworthiness remain obstacles to their coordinated deployment. This study investigates the relationships among the principal technologies underlying industrial intelligence, with particular attention to their functional dependencies, implementation constraints, and capacity to support reliable industrial operations. A structured literature review was conducted using a corpus comprising 55 peer-reviewed publications and 29 supplementary technical and institutional sources. Industrial software, the industrial Internet, digital twins and digital threads, and industrial AI were examined through a common framework covering seven dimensions of technological and engineering readiness. The comparative analysis identified two recurring categories of deployment constraints. Industrial software and the industrial Internet were primarily constrained by the accumulation of domain-specific knowledge, legacy-system compatibility, and heterogeneous data interoperability. For digital twins and industrial AI, the review identified additional challenges concerning model verification, uncertainty assessment, and the integration of intelligent decisions into industrial processes. The analysis further identified interdependent requirements for data consistency, mechanism-based modeling, computational resources, and closed-loop decision execution. These findings indicate that the engineering readiness of industrial intelligence depends not only on the maturity of individual technologies but also on the reliability of their interactions within integrated production systems. The study establishes a system-level perspective for assessing industrial intelligence technologies and provides a structured basis for identifying integration bottlenecks, evaluating engineering trustworthiness, and directing research toward verifiable industrial decision-making.

Keywords: Industrial intelligence; Industrial software; Industrial Internet; Digital twins; System integration; Engineering trustworthiness; Intelligent decision-making

1 Introduction

Industrial intelligence increasingly relies on the coordination of engineering software, industrial communication networks, digital representations of physical assets, and artificial intelligence (AI). These technologies support condition monitoring, fault diagnosis, predictive maintenance, and production decision-making, but their integration introduces requirements that extend beyond the performance of individual components. Industrial processes operate under physical, temporal, and safety constraints, while the information used to support intelligent decisions is often distributed across equipment, software platforms, and organizational systems. The central challenge is therefore not simply to improve computational capability, but to ensure that industrial data, engineering models, and intelligent decisions remain consistent throughout operation.

Research on Industry 4.0 has established the importance of interconnected technologies for intelligent manufacturing [1]. Studies of the industrial Internet of Things (IIoT) have examined communication architectures, cyber-physical integration, and interoperability requirements [2–4]. In parallel, digital-twin research has addressed the relationship between physical assets and computational representations, including model synchronization, enabling technologies, and industrial applications [5–7]. Machine learning has also become an established research

direction in manufacturing, supporting process analysis, equipment monitoring, and predictive decision-making [8–10]. These developments provide the technological foundations of industrial intelligence, although they have largely been investigated within separate engineering and computational domains.

The separation is understandable because the technologies perform different functions. Industrial software represents physical mechanisms, engineering requirements, and operational procedures. Industrial communication infrastructures connect equipment and information systems, making operational measurements available for monitoring and analysis. Digital twins combine engineering representations with information about physical assets, while digital threads preserve relationships among information generated across the product and production lifecycle. Industrial AI uses these resources to generate predictions, diagnoses, and recommendations. Their functions are complementary, but the requirements governing their integration are not necessarily compatible.

Predictive maintenance illustrates this difficulty. A diagnostic model may perform well on a benchmark dataset while depending on measurements whose quality, timing, and interpretation differ from those encountered during deployment. Its predictions may also rely on equipment configurations or operating conditions that change over time. Even when a prediction is accurate, the corresponding maintenance decision must remain compatible with production schedules, equipment restrictions, and operational responsibilities. Similar dependencies arise in process optimization and intelligent production control, where computational outputs must be evaluated against physical constraints before they can influence industrial operations. Consequently, algorithmic performance alone provides insufficient evidence of system-level readiness.

Recent developments in industrial foundation models and large language models have extended interest in AI beyond individual prediction tasks toward more general engineering assistance and decision support [11, 12]. At the same time, physics-informed machine learning has provided methods for incorporating governing equations and physical knowledge into computational models [13]. Probabilistic digital-twin research has examined how model representations and operational observations can be combined while accounting for uncertainty [14]. These approaches offer different means of connecting engineering knowledge with data-driven analysis. Nevertheless, incorporating physical information or increasing model capability does not automatically establish reliability under changing industrial conditions. Model validity, information consistency, uncertainty, and the consequences of incorrect decisions remain application-dependent concerns.

A related difficulty concerns the assessment of technological maturity. Established industrial software and communication technologies have accumulated substantial deployment experience, yet their integration can remain constrained by proprietary interfaces, legacy equipment, and heterogeneous information structures. Digital twins and industrial AI have demonstrated capabilities in selected applications, but their wider operational use introduces additional questions concerning model fidelity, robustness, verification, and decision traceability. The distinction is important: a technology may be mature within its own application domain without being sufficiently integrated or verified for a particular industrial intelligence system. Evaluating technologies independently therefore provides only a partial account of their engineering readiness.

Existing reviews offer detailed accounts of individual technological domains, but their evaluation criteria differ. Industrial networking research emphasizes communication performance and interoperability; digital-twin studies examine representation, synchronization, and model fidelity; and industrial AI research commonly focuses on predictive performance, generalization, and computational efficiency. These criteria are appropriate within their respective fields, yet they do not provide a common basis for examining how limitations propagate between connected technologies. The unresolved issue is how technological capabilities, system dependencies, and engineering trustworthiness can be assessed together without reducing their different requirements to an artificial numerical ranking.

This review addresses that issue through a structured comparison of four technological domains: industrial software, the industrial Internet, digital twins and digital threads, and industrial AI. Seven dimensions are used to examine their engineering characteristics: technological maturity, data dependence, model interpretability, real-time requirements, system integration effort, engineering trustworthiness, and barriers to large-scale deployment. The analysis also considers four forms of system dependence involving industrial data, engineering models, computational resources, and decision execution.

The contribution is threefold. First, the review places the four technological domains within a common system-oriented framework, clarifying their functional roles and interdependencies. Second, it distinguishes legacy-integration constraints from trustworthiness-verification constraints, providing a basis for interpreting differences between technological maturity and operational readiness. Third, it connects these findings to research priorities in semantic interoperability, mechanism–data integration, system-level verification, and trustworthy industrial decision-making. The resulting framework is intended for qualitative comparison and the identification of engineering requirements rather than numerical ranking or experimental validation of an integrated system.

Section 2 describes the literature selection procedure and analytical framework. Section 3 examines the four technological domains and their system-level dependencies. Section 4 discusses engineering trustworthiness,

deployment constraints, and research priorities. Section 5 presents the conclusions.

2 Review Methodology and Analytical Framework

The review examined industrial intelligence from a system-level perspective, focusing on the relationship between technological capabilities and the engineering conditions required for reliable operation. Three questions guided the analysis: What functions do the principal technologies perform within industrial intelligence systems? How do their operational characteristics and integration requirements differ? Which dependencies and engineering constraints affect their trustworthiness? To address these questions, the review combined structured literature selection with a qualitative comparison of four technological domains.

2.1 Literature Search and Source Selection

The principal academic literature search was conducted in September 2025, covering publications available up to that date. Earlier foundational studies were identified through supplementary searches and backward citation tracing. Technical and institutional sources were considered separately, with subsequent updates incorporated where relevant to the discussion of industrial platforms and standards. Web of Science Core Collection, Scopus, and Engineering Village served as the principal bibliographic databases. Institute of Electrical and Electronics Engineers (IEEE) Xplore and ScienceDirect were additionally consulted for methodological research concerning physics-informed learning, neural operators, surrogate modeling, and related computational approaches.

The search combined general terms associated with industrial intelligence, smart manufacturing, and Industry 4.0 with expressions referring to industrial software, the industrial Internet, digital twins, digital threads, and industrial AI. Review-oriented terms, including review, survey, framework, and challenges, were used to identify publications offering broader assessments of technological development. The search sources, principal terms, and reported retrieval counts are summarized in Table 1.

Table 1. Literature search strategy and reported retrieval records

Source	Search Focus	Main Search Terms or Approach	Records
Web of Science Core Collection	Industrial intelligence and enabling technologies	Intelligent industry, smart manufacturing, Industry 4.0, industrial Internet, digital twin, digital thread, industrial AI; combined with review-oriented terms	301
Scopus	Industrial intelligence and enabling technologies	Corresponding title, abstract, and keyword search using the principal technology terms and review-oriented expressions	332
Engineering Village	Industrial technologies and engineering applications	Technology-related terms combined with review, survey, and framework expressions	248
IEEE Xplore and ScienceDirect	Mechanism–data integration	Physics-informed learning, neural operators, surrogate modeling, scientific machine learning, and related industrial applications	86
Backward citation searching	Foundational and methodological literature	References traced from relevant review publications	45
Total identified			1,012
After deduplication			588

Note: AI = artificial intelligence; IEEE = Institute of Electrical and Electronics Engineers.

The literature identification and screening process is presented in Figure 1. The search initially identified 1,012 records, of which 588 remained after duplicate removal. Title and abstract screening excluded 440 records, leaving 148 publications for full-text assessment. A further 93 publications were excluded because they were outside the research scope (34), examined generic algorithms without identifiable industrial applications (26), relied on secondary information without traceable original sources (19), or had unavailable or non-English full texts (14). The final academic corpus comprised 55 peer-reviewed publications.

In parallel, 29 supplementary sources were collected separately, comprising 12 market and technology reports, eight standards documents, and nine technical white papers or product documents. These sources provided contextual information concerning industrial platforms, standardization activities, and technological developments. They were not treated as equivalent to peer-reviewed evidence when assessing model performance or engineering reliability. The included academic publications and supplementary sources were recorded separately according to publication type and their role in the review. This classification was used to distinguish peer-reviewed evidence from technical descriptions, standards, and industry reports during the qualitative synthesis.

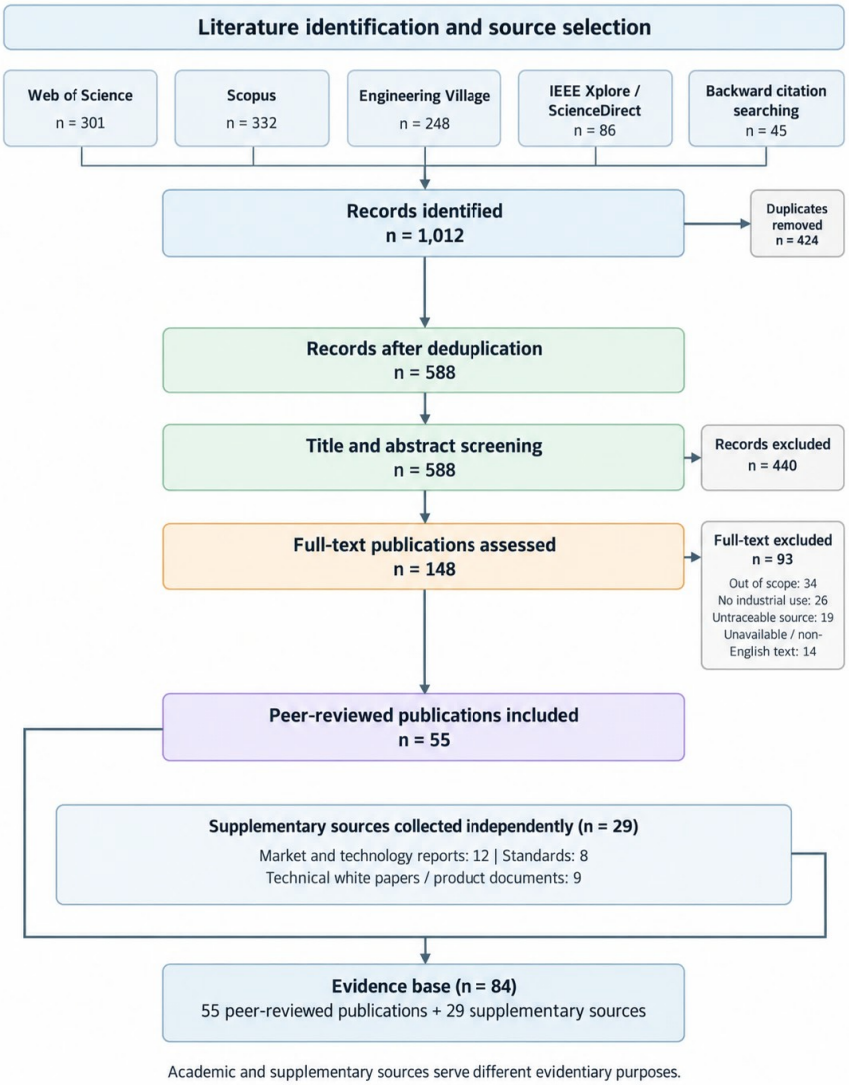


Figure 1. Flow diagram of literature identification, screening, eligibility assessment, and source inclusion

Note: IEEE = Institute of Electrical and Electronics Engineers.

Publications were considered eligible when they examined the technological foundations, industrial applications, operational requirements, or engineering limitations of at least one of the four domains. Particular attention was given to studies addressing system integration, interoperability, physical-model consistency, uncertainty, and reliable industrial decision-making. Evidence was interpreted in relation to each publication's application conditions and reported limitations rather than its publication venue alone.

2.2 Evidence Organization and Interpretation

The selected literature was organized into four technological domains: industrial software, the industrial Internet, digital twins and digital threads, and industrial AI. This division reflected their principal engineering functions without assuming that they operated independently. Industrial software was examined in relation to engineering models, process knowledge, simulation, and lifecycle information management. The industrial Internet was considered through communication architecture, operational data acquisition, and interoperability. Digital twins and digital threads were assessed in terms of physical–digital representation, model synchronization, and information continuity, while industrial AI was examined through predictive modeling, diagnosis, optimization, and decision support.

The synthesis distinguished evidence concerning established technological capabilities, operational requirements, and unresolved engineering constraints. This distinction was necessary because the technologies are commonly evaluated using different performance measures. Communication latency, for example, is central to industrial networking, whereas model fidelity is more relevant to digital twins and predictive accuracy to industrial AI. Such measures were interpreted within their original application conditions rather than combined into a single numerical indicator.

The comparison also distinguished demonstrated capabilities from broader claims of industrial applicability. A successful application in a particular production environment does not necessarily establish transferability to other equipment or operating conditions. Similarly, favorable benchmark performance does not independently demonstrate reliability under industrial disturbances or changing data distributions. These differences informed the subsequent assessment of technological maturity and engineering trustworthiness.

2.3 Cross-Technology Analytical Framework

A common analytical framework was used to compare technologies whose engineering functions, operational requirements, and evaluation criteria differ substantially. As illustrated in Figure 2, the framework connects four technological domains with seven dimensions of assessment and examines the dependencies and constraints that arise during their integration into industrial intelligence systems.

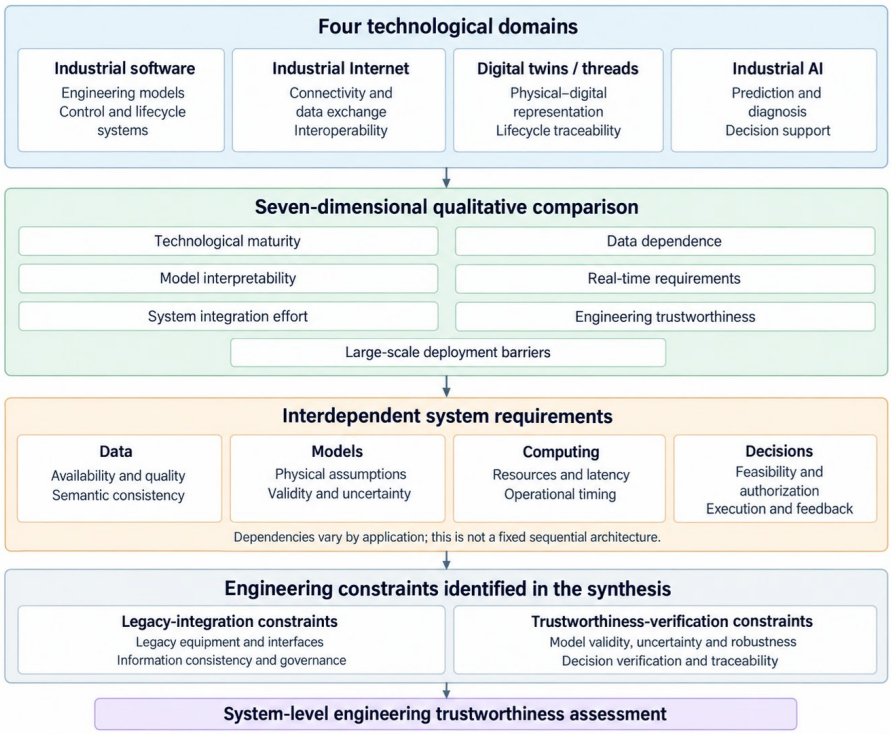


Figure 2. Analytical framework for comparing industrial intelligence technologies, system-level dependencies, and engineering constraints

Note: AI = artificial intelligence.

The seven dimensions comprise technological maturity, data dependence, model interpretability, real-time requirements, system integration effort, engineering trustworthiness, and barriers to large-scale deployment. Their definitions and principal assessment considerations are presented in Table 2.

Technological maturity was examined through established applications, deployment experience, and standardization. Data dependence concerned information availability, quality, continuity, and semantic consistency. Model interpretability addressed the extent to which computational behavior could be related to identifiable engineering principles or decision criteria. Real-time requirements concerned operational timing constraints, while system integration effort reflected the difficulty of connecting technologies with existing equipment, software, and information structures.

Engineering trustworthiness was distinguished from technological maturity. It concerned the evidence supporting model validity, operational robustness, uncertainty assessment, traceability, and reliable performance under intended engineering conditions. Barriers to large-scale deployment included legacy-system compatibility, computational requirements, data governance, maintenance, and engineering verification.

The seven dimensions were used for qualitative comparison rather than numerical scoring. No common weights or maturity index were assigned because their relative importance depends on the industrial application. Real-time requirements may dominate a control application, for example, whereas information continuity and configuration management may be more consequential for digital-thread implementation. The framework therefore supports comparison without assuming that different technological domains can be assessed against identical performance thresholds.

Table 2. Analytical dimensions for the cross-technology assessment

Dimension	Assessment Focus	Principal Evidence Considered
Technological maturity	Development and industrial adoption	Established applications, deployment experience, standardization and reported technological limitations
Data dependence	Information required for reliable operation	Data availability, quality, continuity, semantic consistency and domain specificity
Model interpretability	Accessibility of model behavior and reasoning	Physical representation, explicit rules, model transparency and traceability
Real-time requirements	Compatibility with operational timing constraints	Communication latency, synchronization, computational delay and control-cycle requirements
System integration effort	Difficulty of incorporating the technology into existing industrial systems	Interface compatibility, legacy equipment, data conversion, software integration and organizational coordination
Engineering trustworthiness	Evidence supporting reliable operation under intended conditions	Verification, uncertainty assessment, robustness, reliability and safety-related constraints
Barriers to large-scale deployment	Constraints on extending applications beyond limited installations	Implementation cost, scalability, governance, maintenance and engineering validation

2.4 System Dependencies and Synthesis Procedure

The final stage examined how technological capabilities and limitations interact within integrated industrial intelligence systems. Four forms of dependency were considered: data, models, computational resources, and decision execution. These dependencies describe functional relationships rather than a fixed sequence of technological layers.

Data dependence concerns the availability, timing, quality, and semantic consistency of operational information. Industrial communication infrastructures make measurements accessible to analytical systems, but connectivity alone does not ensure that information from different equipment or software platforms can be interpreted consistently. Model dependence concerns the engineering representations required to interpret these measurements and evaluate physical behavior. Industrial software supplies mechanism-based models, process constraints, and lifecycle information, while digital twins and AI use these resources for simulation, prediction, and decision support.

Computational dependence concerns the resources required for simulation, learning, inference, and optimization within operational time constraints. High-fidelity models may provide detailed physical representations but require computational resources incompatible with time-sensitive applications. Decision-execution dependence concerns the interfaces and procedures through which analytical outputs are evaluated, authorized, and translated into industrial actions. A prediction or optimization result acquires operational significance only when its implementation is compatible with equipment capabilities, process constraints, and decision responsibilities.

The comparison examined how limitations associated with these dependencies affect industrial deployment. Two recurring categories of constraints were distinguished. Legacy-integration constraints concern established equipment, proprietary interfaces, heterogeneous information structures, and accumulated engineering knowledge. Trustworthiness-verification constraints concern model validity, uncertainty, robustness, and the verification of intelligent decisions under changing operating conditions.

These categories were developed by comparing the engineering difficulties discussed across the four technological domains. They were interpretive rather than statistically derived and were not treated as mutually exclusive. An industrial system may encounter legacy integration problems while also requiring extensive verification of its computational models and intelligent functions. The distinction provides a basis for examining why established technologies and emerging intelligent methods face different, sometimes overlapping, deployment difficulties.

The resulting synthesis focused on the relationship between technological capabilities, system dependencies, and engineering trustworthiness. It did not seek to rank technologies numerically or infer universal performance differences. Instead, the framework was used to identify recurring engineering constraints and examine how their interactions affect the readiness of integrated industrial intelligence systems. Section 3 applies this perspective to the four technological domains and their system-level relationships.

3 Key Technologies and Their System-Level Dependencies

Industrial intelligence depends on technologies that perform different but complementary engineering functions. Industrial software represents physical mechanisms, engineering requirements, and operational procedures. The

industrial Internet connects equipment and information systems, enabling the acquisition and exchange of production data. Digital twins and digital threads link physical assets with computational representations and lifecycle information, while industrial AI supports prediction, diagnosis, optimization, and decision-making. These functions are interconnected, but their integration does not follow a universal architectural sequence. The relevant relationships depend on the industrial process, available infrastructure, and requirements of the intended application.

The following analysis examines the established capabilities and engineering limitations of each technological domain. Particular attention is given to the conditions under which individual capabilities can contribute to reliable industrial intelligence, providing the basis for the cross-technology comparison in Section 3.5.

3.1 Industrial Software and Mechanism-Based Modeling

Industrial software provides the engineering representations through which physical processes, production resources, and operational procedures are modeled and managed. Computer-aided design (CAD), computer-aided engineering (CAE), and electronic design automation (EDA) support product development and engineering analysis. Programmable logic controllers (PLCs), supervisory control and data acquisition (SCADA) systems, and manufacturing execution systems (MES) govern or coordinate production activities, while enterprise resource planning (ERP) and product lifecycle management (PLM) systems organize information across manufacturing and organizational processes.

The importance of industrial software extends beyond the execution of computational tasks. Geometric models, conservation equations, constitutive relationships, process constraints, and equipment operating rules represent engineering knowledge accumulated through design, experimentation, and industrial practice. These representations establish the physical and procedural conditions under which subsequent analysis and decision-making can be performed. Model-based systems engineering (MBSE) provides a structured means of connecting system requirements, architecture, behavior, and verification activities within engineering models [15, 16].

Open-source environments, including OpenFOAM, FEniCS/DOLFINx, and SU2, provide numerical modeling and simulation capabilities for different classes of engineering problems [17–19]. Their availability broadens access to computational methods, but model reliability remains dependent on the governing assumptions, boundary conditions, numerical accuracy, and validation procedures. A computational result cannot be considered reliable solely because it was obtained using established software.

Industrial software development is also shaped by the complexity of maintaining engineering knowledge and compatibility across platforms. Commercial engineering systems commonly integrate modeling kernels, simulation tools, design databases, and lifecycle management functions developed over extended periods. Industry assessments and corporate developments illustrate the commercial significance of these capabilities [20–23]. However, market position and software adoption do not independently establish engineering reliability. For industrial intelligence, the more consequential issue is whether information represented in different software environments can be exchanged and interpreted consistently. Earlier industrial analytics initiatives illustrate efforts to connect equipment data with enterprise analytics [24].

Recent developments have introduced generative AI into industrial programming, engineering assistance, and enterprise information systems. Siemens Industrial Copilot and SAP Joule illustrate the incorporation of language-model-based functions into industrial and enterprise workflows [25, 26]. Related industrial partnerships illustrate continued development of AI-assisted engineering tools [27], while broader assessments discuss the economic context of generative AI [28]. These systems can assist with information retrieval, programming, and technical documentation, but generated outputs remain subject to engineering verification. In production environments, a plausible program or recommendation cannot substitute for compliance with equipment capabilities, operational procedures, and safety requirements.

A further limitation concerns the computational demands of mechanism-based modeling. High-fidelity simulations can represent complex physical behavior but may require execution times incompatible with online monitoring or operational decision-making. Reduced-order models, learned surrogates, and physics-informed methods provide alternatives for selected applications [13]. Their usefulness depends on approximation accuracy, operating-domain coverage, and the treatment of uncertainty. Industrial software therefore contributes both engineering knowledge and operational interfaces to industrial intelligence, while its principal integration constraints concern proprietary data structures, legacy compatibility, computational requirements, and the continuing validity of engineering models.

3.2 Industrial Internet and Data Interoperability

The industrial Internet provides the communication infrastructure connecting production equipment, controllers, computing resources, and enterprise applications. Its requirements differ from those of general-purpose networks because industrial processes may depend on predictable response times, continuous availability, and reliable

coordination between sensing and actuation. Communication systems must therefore support information exchange while accommodating the operational constraints of physical production processes.

Industrial communication has developed through established automation networks and the increasing integration of information technology (IT) with operational technology (OT). Research on industrial networking has examined the transition toward interconnected automation architectures, while studies of IIoT have identified the opportunities and limitations associated with industrial connectivity [29, 30]. These developments have made operational information more accessible, but connectivity alone does not establish interoperability.

Industrial networks incorporate protocols such as Modbus, PROFINET, and EtherNet/IP, reflecting differences in equipment generations and automation architectures. Gateways can enable communication between otherwise incompatible systems, yet transmitted information may still differ in measurement units, sampling intervals, equipment identifiers, and interpretations of operating states. Such differences become consequential when data from multiple systems are combined for diagnosis, simulation, or intelligent decision-making. Semantic interoperability is therefore distinct from the ability to establish a communication connection.

Open Platform Communications Unified Architecture (OPC UA) provides mechanisms for structured information exchange and interoperable industrial communication. Developments involving time-sensitive networking further address the need for predictable communication behavior in selected industrial applications [31]. Nevertheless, standards adoption remains affected by legacy equipment, implementation cost, and the information structures already used within production systems. The reliability of integrated industrial intelligence depends on the consistency of these structures as well as the communication mechanisms connecting them.

The distribution of computational functions introduces another architectural consideration. Edge computing supports local data processing, monitoring, and inference, while centralized environments provide resources for historical analysis and computationally intensive operations [32, 33]. Industry forecasts provide contextual evidence of investment interest in IoT and edge computing, rather than independently verified engineering performance [34, 35]. The allocation of functions between equipment, edge nodes, and centralized platforms must reflect the timing and reliability requirements of the application. A configuration suitable for production reporting may not satisfy the constraints of real-time control.

These requirements become particularly important when industrial data are used for AI. Models developed using cleaned and synchronized historical datasets may encounter missing observations, delayed messages, sensor replacement, or changes in operating conditions during deployment. Such changes can affect predictions even when the model itself remains computationally functional. Data quality and semantic consistency must consequently be maintained throughout operation rather than treated as isolated preprocessing tasks.

Connectivity also introduces cybersecurity concerns. The convergence of operational and IT increases the exposure of production systems to unauthorized access, data manipulation, and service disruption. Industrial cybersecurity requires coordinated consideration of access control, communication integrity, system availability, and production continuity. International Electrotechnical Commission (IEC) 62443-3-3 specifies system security requirements and security levels for industrial automation and control systems [36]. Its relevance to industrial intelligence lies in the need to protect interconnected functions without compromising the operational requirements of the physical process.

The industrial Internet provides the information infrastructure and communication pathways required for integrated industrial intelligence. Industrial automation partnerships also illustrate efforts to combine operational platforms with AI computing resources [37]. Nevertheless, the integration of industrial networks and intelligent functions remains constrained by heterogeneous equipment, semantic interoperability, timing requirements, cybersecurity, and the continuing integrity of operational data.

3.3 Digital Twins and Digital Thread Integration

Digital twins provide computational representations of physical assets, processes, or systems that are connected to information about their real-world counterparts. Their defining characteristic is the maintained relationship between a physical system and its computational representation. Depending on the intended application, this relationship may involve state synchronization, simulation, parameter updating, condition assessment, or prediction of future behavior.

Digital-twin research has examined the concept from modeling, architectural, and manufacturing perspectives [38–42]. Additional studies address enabling technologies, industrial applications, and digital twins in cyber-physical production systems [43–47]. These studies distinguish digital twins from static digital models by emphasizing the role of physical-system information and the functions supported through physical–digital interaction. However, the required degree of synchronization and model fidelity varies substantially between applications.

An equipment-monitoring twin may require frequent updates of a limited number of operating variables, whereas a twin supporting structural analysis or process optimization may depend on detailed physical models. High-fidelity representations can improve the description of relevant physical mechanisms, but their computational demands may

restrict online use. Reduced-order and data-driven approximations can address this difficulty, although their reliability depends on the operating conditions represented during model development. Probabilistic predictive digital twins provide a basis for combining physical representations with observations while accounting for uncertainty [14].

The digital thread addresses a related but distinct requirement. Whereas a digital twin represents and analyzes a physical system, a digital thread preserves relationships among engineering and operational information across lifecycle stages. Design specifications, manufacturing plans, production records, equipment configurations, and maintenance histories may be distributed across different software environments. Maintaining their relationships is important when decisions depend on changes extending beyond current sensor measurements.

Digital-thread integration becomes particularly relevant when equipment configurations evolve. A predictive model may remain associated with an earlier asset configuration, while the physical equipment has undergone maintenance or modification. Without traceable configuration information, the computational representation may no longer describe the operating system accurately. Research on digital-thread-based engineering has examined how lifecycle information can support continuity and traceability in engineering design [48].

Manufacturing standards and engineering information frameworks contribute to the organization of these relationships. ISO 23247-1 establishes general principles for digital-twin frameworks in manufacturing [49], while MBSE-oriented research has examined the use of digital twins to connect engineering models and system development activities [50]. Commercial developments involving industrial simulation and visualization platforms also illustrate the growing integration of physical modeling, operational information, and computing resources [51–53]. Research on industrial metaverse architectures and language-model-assisted digital-twin modeling describes further directions for industrial information interaction [54, 55]. These developments illustrate progress in industrial modeling and platform integration, but they do not establish model validity or operational reliability across different deployment conditions.

The engineering value of a digital twin depends on its ability to represent the physical system sufficiently well for the intended task. Research on digital-twin modeling has examined the trade-offs among model fidelity, computational cost, and the representation of physical-system behavior [56]. Model calibration, data synchronization, configuration management, and uncertainty assessment are therefore continuing requirements. A twin developed for monitoring may not possess the fidelity needed for optimization, while a detailed simulation may be unsuitable for time-sensitive decisions. Digital twins and digital threads consequently provide an important connection between industrial software, operational data, and AI, but their trustworthiness depends on the consistency and verification of these relationships.

3.4 Industrial Artificial Intelligence

Industrial AI applies computational learning and decision-support methods to manufacturing processes governed by physical and operational constraints. Its applications include equipment fault diagnosis, predictive maintenance, quality inspection, process monitoring, production scheduling, and optimization. Unlike general-purpose predictive modeling, industrial applications require outputs that remain meaningful under changing operating conditions and can be incorporated into established engineering procedures.

The methodological development of industrial AI extends from rule-based systems and conventional machine learning to deep learning and more general computational models. Rule-based approaches provide explicit decision logic but may be difficult to maintain in complex environments. Statistical learning methods infer relationships from operational observations, while deep learning supports the analysis of industrial images, signals, and multivariate time series. Recent research on industrial foundation models and large language models has expanded interest toward engineering assistance, heterogeneous information processing, and intelligent interaction with software tools [11, 12]. Broader AI development reports provide context on adoption trends, but not direct evidence of industrial reliability [57].

Predictive maintenance illustrates the relationship between algorithmic capability and industrial applicability. Machine-learning methods have been developed for fault classification, condition assessment, and remaining useful life estimation. Reviews of machine-learning-based fault diagnosis and predictive maintenance have examined their methodological development, industrial applications, and implementation challenges [58, 59]. Automotive predictive-maintenance research provides further evidence of deployment-related difficulties [60]. Public datasets such as National Aeronautics and Space Administration (NASA)'s C-MAPSS, the Case Western Reserve University (CWRU) bearing dataset, and MIMII provide established resources for evaluating selected diagnostic and prognostic tasks [61–63]. These datasets support reproducible comparisons under defined conditions, but their results do not establish equivalent performance across different equipment populations or production environments.

Industrial data introduce difficulties that are less prominent in many conventional learning tasks. Normal operating observations may be abundant, while examples of critical failures are scarce. Equipment behavior changes with load, environmental conditions, maintenance interventions, and operating regimes. Models may consequently encounter data distributions that differ from those represented during training. Industrial anomaly

detection research and inspection benchmarks provide methods and evaluation resources for addressing selected aspects of these problems [64, 65]. Nevertheless, robustness under unfamiliar conditions remains dependent on the representativeness of the available data and the characteristics of the intended application.

These difficulties distinguish predictive performance from engineering trustworthiness. Accuracy, recall, and related measures describe performance under specified evaluation conditions but do not independently establish physical consistency, robustness, or the consequences of incorrect decisions. A fault classifier may achieve high accuracy while remaining sensitive to sensor changes, and a remaining useful life model may exhibit low average error without providing sufficiently reliable estimates near a critical failure threshold. Industrial applications therefore require evaluation criteria related to the decisions supported by the model.

Interpretability and accountability form part of this requirement. In fault diagnosis, engineers may need to determine whether a prediction corresponds to a plausible failure mechanism. In process optimization, the relationship between recommended operating conditions and physical constraints may be more important than the explanation of individual model parameters. Research on interpretable and trustworthy AI has emphasized the importance of examining model behavior, robustness, and decision accountability [66–68]. For industrial intelligence, these properties must be evaluated in relation to the physical process and operational consequences.

Mechanism–data integration provides a methodological connection between industrial AI and established engineering knowledge. Physics-informed neural networks incorporate governing equations or physical constraints into learning, while broader scientific machine-learning approaches combine numerical modeling with data-driven methods [13, 69–73]. Neural operators approximate mappings between functions or parameter fields and their corresponding solutions, potentially reducing the cost of repeated simulation [74, 75]. Surrogate and multifidelity methods provide alternatives when high-fidelity computation is too expensive for repeated evaluation [76]. Knowledge-guided learning incorporates domain relationships or explicit engineering knowledge into statistical modeling [77].

These methods address different limitations and cannot be treated as interchangeable solutions. Physics-informed learning depends on the adequacy of the governing relationships and the numerical behavior of the learning procedure. Neural operators and surrogate models require assessment of generalization beyond the conditions represented during training. Knowledge-guided methods depend on the accuracy and maintenance of the encoded engineering knowledge. Incorporating physical information can restrict implausible predictions, but it does not eliminate uncertainty or the need for application-specific verification.

The emergence of industrial agents introduces additional requirements when computational outputs influence engineering actions. Language-model-based systems can assist with programming, technical documentation, and information retrieval, but generated instructions must be checked against equipment capabilities and operational constraints before execution. A distinction is therefore required between advisory functions, supervised decision support, and actions that directly affect physical processes.

Industrial AI depends on operational measurements supplied through industrial networks, engineering representations maintained within industrial software, and, where appropriate, simulation or state-estimation functions provided by digital twins. Its predictions and recommendations must subsequently be interpreted through maintenance, manufacturing execution, or control systems. The reliability of this process depends on the validity of the analytical model and the consistency of the information and interfaces connecting it to industrial operations.

3.5 Cross-Technology Dependencies and Comparative Assessment

The preceding analysis shows that industrial software, the industrial Internet, digital twins, and industrial AI contribute distinct capabilities to manufacturing systems. Their individual development, however, provides only a partial account of industrial intelligence readiness. Integrated operation depends on relationships among engineering representations, operational information, computational resources, and decision processes. These relationships introduce requirements that cannot be assessed adequately through the performance measures of any single technological domain.

Four forms of dependency are particularly important. Data dependence concerns the availability, quality, timing, and semantic consistency of operational information. Model dependence concerns the engineering representations required to interpret measurements and evaluate physical behavior. Computational dependence concerns the resources needed for simulation, learning, inference, and optimization within operational time constraints. Decision-execution dependence concerns the interfaces and procedures through which analytical outputs are verified, authorized, and implemented.

These dependencies are functional rather than strictly sequential. Industrial software may operate directly at the equipment level, AI may be deployed at the edge, and digital-twin functions may be distributed across several computing environments. Their interaction depends on the industrial application rather than a universal technological hierarchy.

The relationships among these components are illustrated in Figure 3. Industrial data and engineering models

provide information for digital-twin functions and AI, while predictions and candidate decisions must be evaluated against physical and operational constraints before implementation. Execution outcomes return to the physical process and can subsequently be observed through the data infrastructure. This feedback relationship illustrates how limitations in information consistency, model validity, or decision execution can affect the reliability of the integrated system.

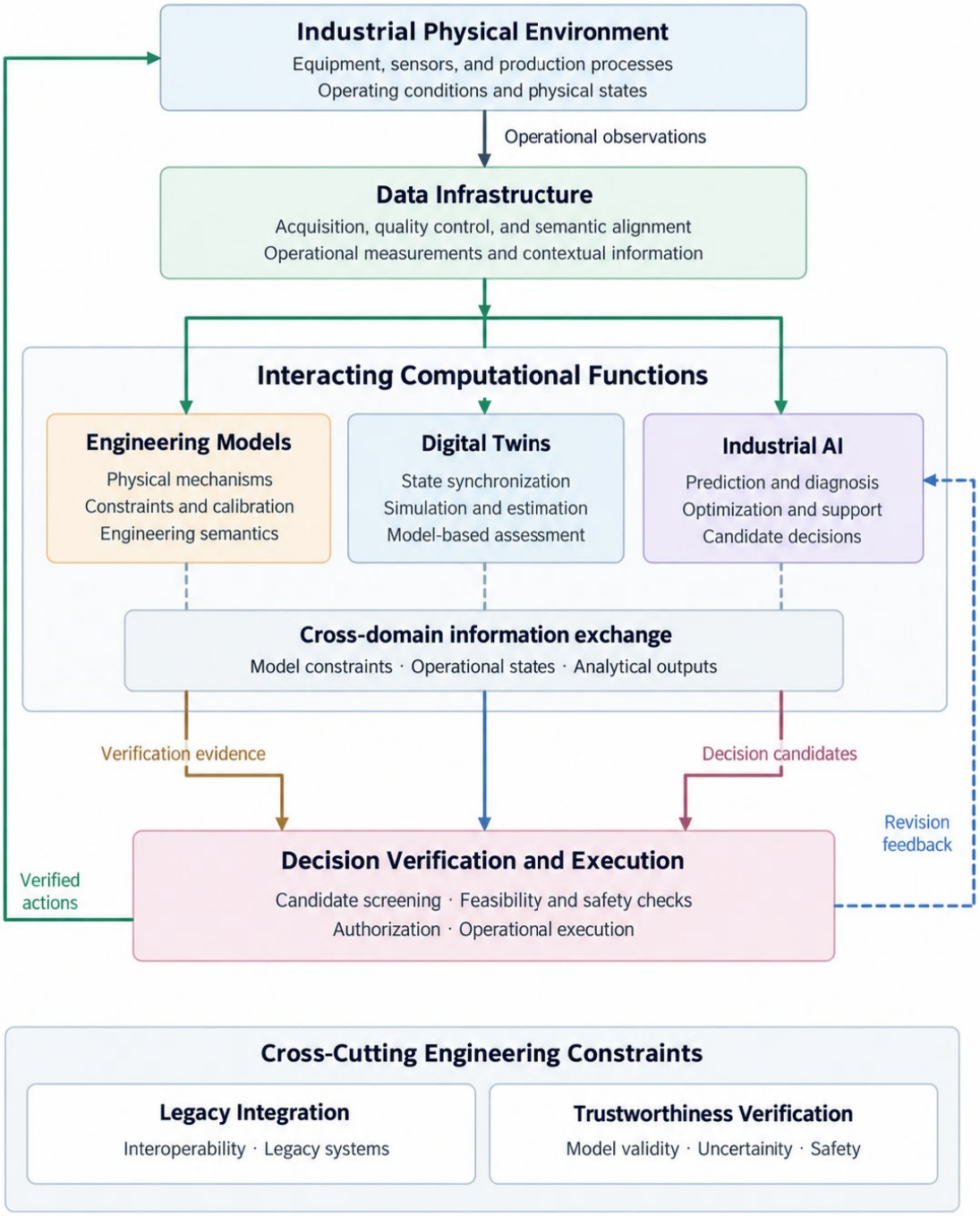


Figure 3. System-level dependencies and engineering constraints affecting the trustworthiness of integrated industrial intelligence systems

Note: AI = artificial intelligence.

The seven-dimensional framework introduced in Section 2 was applied to compare the four technological domains. Table 3 summarizes their characteristics in terms of technological maturity, data dependence, model interpretability, real-time requirements, system integration effort, engineering trustworthiness, and barriers to large-scale deployment. The entries represent a qualitative synthesis of the reviewed evidence rather than numerical maturity scores or statistically estimated differences.

Table 3. Cross-technology comparison of industrial intelligence technologies

Comparison Dimension	Industrial Software	Industrial Internet	Digital Twins and Digital Threads	Industrial AI
Technological maturity	Established engineering applications and mature software ecosystems; capabilities vary by function	Extensive industrial deployment with established communication technologies and evolving interoperability standards	Established applications in selected domains; broader lifecycle integration remains uneven	Mature applications in selected prediction and diagnostic tasks; autonomous industrial decision-making remains less established
Data dependence	Engineering knowledge, process parameters, configuration records, and lifecycle information	Continuous equipment measurements, communication metadata, and consistent information models	Synchronized operational observations, physical models, and cross-lifecycle information	Representative training and operational data, with particular difficulty obtaining rare fault and abnormal-condition samples
Model interpretability	Explicit engineering models and procedural logic, although some proprietary implementations remain opaque	Primarily concerned with communication and information semantics rather than predictive-model interpretation	Depends on the modeling approach; mechanism-based components may be interpretable, while learned approximations require additional examination	Varies considerably by method; complex learning models may require explanation, physical constraints, or independent verification
Real-time requirements	Application-dependent, ranging from offline engineering simulation to deterministic control functions	Stringent timing and availability requirements in field communication; less restrictive conditions for enterprise analytics	Determined by synchronization frequency, simulation complexity, and intended operational use	Depends on inference and decision functions; advisory analysis and closed-loop applications impose different requirements
System integration effort	Legacy-system compatibility, proprietary interfaces, model exchange, and configuration management	Heterogeneous protocols, equipment generations, semantic interoperability, and OT/IT coordination	Integration of physical models, operational measurements, and lifecycle information across systems	Integration of data pipelines, model execution, engineering constraints, and operational decision processes
Engineering trustworthiness	Supported by established verification practices in many applications; dependent on software function and configuration	Supported by industrial reliability engineering and communication standards; security and interoperability remain important	Dependent on model fidelity, synchronization, uncertainty assessment, and continuing validation	Dependent on data representativeness, robustness, interpretability, uncertainty, and verification of decision consequences
Main barriers to large-scale deployment	Accumulated engineering knowledge, proprietary dependencies, licensing costs, and legacy-system migration	Retrofitting costs, semantic inconsistency, cybersecurity, and data governance	Model construction and maintenance, cross-lifecycle data consistency, computational requirements, and verification	Data scarcity, distribution shift, uncertain model behavior, operational integration, and accountability for decisions

Note: OT = operational technology; IT = information technology; AI = artificial intelligence.

The comparison identifies different patterns of engineering constraints. Industrial software and industrial

communication infrastructures have accumulated substantial deployment experience, but their integration remains affected by proprietary interfaces, legacy equipment, heterogeneous information structures, and the difficulty of maintaining consistent engineering knowledge. In the reviewed applications, these challenges mainly concerned the integration, compatibility, and maintenance of established engineering systems.

Digital twins and industrial AI present a different combination of requirements. Their wider operational use depends on model validity, uncertainty assessment, robustness under changing conditions, and the verification of computational outputs. For digital twins, these requirements concern the correspondence between physical assets and computational representations, together with the continuity of model updating. For industrial AI, they concern data representativeness, model generalization, interpretability where required, and the consequences of intelligent decisions.

These findings support a distinction between legacy-integration constraints and trustworthiness-verification constraints. The former concern the difficulty of connecting, adapting, and maintaining established technological assets. The latter concern the evidence needed to determine whether computational models and intelligent functions can be relied upon under intended operating conditions. The categories are not mutually exclusive. Industrial software may contain learning-based components requiring extensive verification, while digital twins may encounter severe legacy integration problems.

Predictive maintenance provides a practical example of their interaction. Operational measurements are collected through industrial communication infrastructure and associated with equipment identifiers, operating conditions, and maintenance histories. Engineering models may supply information about expected physical behavior, while digital twins and AI support condition assessment and prediction. The resulting maintenance recommendation must then be evaluated within the production and maintenance systems responsible for its implementation.

A limitation at any stage can affect the reliability of the complete process. Inconsistent identifiers may associate measurements with the wrong asset, an outdated model may fail to represent equipment modifications, and a predictive model may perform poorly under operating conditions absent from its training data. Even a reliable prediction may produce an unsuitable recommendation if maintenance constraints or production requirements are ignored. These examples illustrate possible dependency-related failure mechanisms rather than experimentally measured failure rates.

The same relationships arise in production scheduling, process optimization, and quality inspection. Resource information, equipment constraints, model validity, and operational authorization determine whether analytical outputs can be translated into appropriate industrial actions. Faster inference cannot resolve inconsistent data semantics, and greater simulation fidelity does not guarantee that a digital representation reflects the current equipment configuration. Similarly, improved predictive accuracy does not establish that an automated recommendation satisfies the requirements of a production system.

Engineering trustworthiness must therefore be considered at both component and system levels. Component-level assessment concerns properties such as numerical accuracy, communication reliability, model robustness, and compliance with specified engineering requirements. System-level assessment additionally examines whether information remains consistent across interfaces, computational models remain valid under operating conditions, and intelligent decisions can be verified and implemented within the constraints of the physical process. These requirements are closely related but cannot be reduced to a single measure of technological readiness. The comparative analysis indicates that reliable industrial intelligence depends on the coordination of established engineering capabilities with verifiable computational functions. This distinction provides the basis for examining mechanism–data integration, system-level verification, security, and decision accountability in Section 4.

4 Engineering Trustworthiness, Deployment Barriers, and Research Directions

The cross-technology comparison in Section 3 indicates that industrial intelligence readiness depends on more than the maturity of individual technologies. Established software and communication systems face difficulties associated with legacy equipment and heterogeneous information structures, while digital twins and AI introduce additional requirements concerning model validity, uncertainty, and decision verification. These constraints become interdependent when computational functions are incorporated into industrial operations. A reliable predictive model, for example, may produce unsuitable recommendations if its inputs are inconsistent with the current equipment configuration or if the proposed action violates operational constraints.

This section examines the engineering requirements arising from these dependencies and the challenges involved in addressing them within integrated industrial systems. It first considers data governance and semantic interoperability, followed by mechanism–data integration and model trustworthiness, with particular attention to the consistency, reliability, and interpretability of integrated approaches. The discussion then addresses system-level verification, cybersecurity, functional safety, and decision accountability, highlighting their importance in ensuring reliable and secure system operation. Finally, it identifies research priorities for integrated industrial intelligence.

4.1 Data Governance and Semantic Interoperability

Industrial intelligence systems draw on information generated by equipment, controllers, engineering software, MES, and enterprise applications. These sources were developed for different purposes and may represent the same physical asset using different identifiers, measurement conventions, timestamps, or configuration records. Consequently, data exchange does not necessarily ensure that the transferred information can be interpreted consistently.

The difficulty extends beyond missing values or measurement noise. A sensor reading may be numerically correct but unsuitable for an analytical task when its engineering context is incomplete. Temperature measurements, for instance, may require information about sensor location, operating mode, calibration, and equipment configuration. Maintenance records can present similar difficulties when interventions are not consistently linked to the operational data collected before and after equipment modification. Combining such information without adequate contextual alignment may produce computationally consistent but physically misleading representations.

Industrial information standards provide mechanisms for addressing aspects of this problem. OPC UA supports structured information exchange, while manufacturing digital-twin frameworks contribute to the organization of relationships among physical assets and computational representations [31, 49]. Nevertheless, common interfaces cannot resolve every inconsistency in engineering meaning. Existing production systems may retain local naming conventions, proprietary data structures, and configuration practices that are not explicitly represented in exchanged information.

Digital-thread engineering addresses information continuity across lifecycle stages. Design specifications, manufacturing records, equipment configurations, and maintenance histories must remain associated with the correct physical assets as industrial systems evolve [48]. This requirement is particularly important for AI because a model trained using historical observations may depend on equipment conditions that no longer apply during deployment.

The corresponding research problem is to maintain engineering meaning across systems and over time. Semantic models, consistent asset identification, configuration traceability, and application-specific data-quality assessment provide possible directions. Their effectiveness should be examined through industrial information exchanges and operational tasks, rather than inferred solely from compliance with communication interfaces.

4.2 Mechanism–Data Integration and Model Trustworthiness

The reliability of industrial intelligence depends on whether computational models adequately represent the physical and operational relationships relevant to their intended applications. Data-driven methods can identify complex patterns in industrial observations, but their predictions may become unreliable when operating conditions differ from those represented during training. Mechanism-based models provide explicit relationships derived from physical laws and engineering knowledge, although detailed simulation and calibration can be computationally demanding. Mechanism–data integration seeks to combine these complementary capabilities. Reviews of scientific knowledge integration and fluid-mechanics machine learning provide further methodological context [78, 79].

Six methodological approaches are particularly relevant. Physics-informed learning incorporates governing equations or physical constraints into model construction, commonly through residual terms or constrained representations [13, 69–73]. Neural operators learn mappings between functions or parameter fields and corresponding solutions, offering alternatives to repeated numerical simulation within defined problem families [74, 75]. Surrogate and multifidelity methods approximate computationally expensive models using simulations or experimental observations [76]. Knowledge-guided learning incorporates domain relationships or engineering rules into statistical modeling [77]. Hybrid digital twins combine physical representations with operational observations and model updating [14], while constraint-aware decision methods restrict candidate actions according to engineering feasibility requirements.

Figure 4 summarizes these approaches and the principal verification requirements associated with them. The diagram emphasizes the relationship between methodological assumptions and the evidence needed to assess physical consistency, predictive reliability, and operational feasibility. It does not imply that one approach is universally superior or that incorporating physical knowledge automatically establishes model trustworthiness.

The approaches differ in the conditions under which they can be used reliably. Physics-informed methods depend on the adequacy of governing relationships and the numerical behavior of the learning procedure. Neural operators and surrogate models require assessment of generalization across relevant parameters, geometries, and operating conditions. Knowledge-guided approaches depend on the validity and maintenance of encoded engineering rules, while hybrid digital twins must account for uncertainty in physical models and operational observations. Constraint-aware decisions require feasible operating regions and restrictions to be specified correctly.

Table 4 extends the conceptual relationships shown in Figure 4 by comparing the application contexts, methodological limitations, and verification conditions of the six approaches. These distinctions are important because the appropriate verification procedure depends on the role of the model within the industrial system.



Figure 4. Mechanism–data integration approaches and their application-specific engineering verification requirements

Table 4. Application contexts, methodological limitations, and verification conditions of mechanism–data integration approaches

Methodological Approach	Integration of Engineering Knowledge	Representative Applications	Principal Limitations	Verification Requirements
Physics-informed learning	Governing equations and physical constraints incorporated into learning	Forward and inverse problems, physical state estimation	Optimization difficulties, sensitivity to residual weighting, complex multiscale behavior	Physical residual assessment, boundary-condition consistency, numerical accuracy
Neural operators	Learned mappings between parameter fields and solution fields	Repeated simulation, parametric analysis, digital-twin computation	Dependence on training-domain coverage and generalization across conditions	Error assessment across parameter ranges, geometries, and operating conditions

Surrogate and multifidelity modeling	Approximation of high-fidelity simulations using lower-cost models	Engineering optimization, uncertainty analysis, repeated simulation	Approximation error and unreliable extrapolation	Comparison with reference models, uncertainty assessment, operating-domain validation
Knowledge-guided learning	Engineering rules, domain relationships, or structured knowledge incorporated into learning	Process monitoring, fault diagnosis, knowledge-intensive decisions	Knowledge formalization, inconsistency, and maintenance requirements	Rule consistency, knowledge validity, traceability of model outputs
Hybrid digital twins	Physical models combined with operational data and probabilistic updating	Condition monitoring, predictive maintenance, state estimation	Model calibration, data synchronization, uncertainty propagation	Model–asset consistency, predictive validation, uncertainty calibration
Constraint-aware decision-making	Physical and operational constraints incorporated into optimization or decision selection	Process optimization, production scheduling, operational decision support	Constraint specification, computational complexity, integration with control systems	Feasibility checking, operational validation, decision traceability

The inclusion of engineering knowledge does not eliminate the need for independent assessment. A physics-informed model may satisfy selected equation residuals while remaining inaccurate because of inappropriate assumptions or incomplete boundary conditions. A surrogate may reproduce a reference simulation within its training domain but become unreliable under unfamiliar conditions. Similarly, a knowledge-guided model may produce logically consistent outputs while relying on outdated process rules.

Engineering trustworthiness therefore requires more than the incorporation of physical information into learning. Model assumptions, uncertainty, operating-domain coverage, and the consequences of prediction errors must be examined in relation to the intended application. Mechanism–data integration is most useful when it establishes a verifiable connection between computational outputs and the engineering requirements governing their use.

4.3 Component-Level Assessment and System-Level Verification

Industrial AI is commonly evaluated using predictive accuracy, classification performance, and computational efficiency. These measures support comparisons under defined experimental conditions but do not fully describe the reliability of an integrated industrial intelligence system. Once a model interacts with operational databases, engineering software, digital twins, and production equipment, its behavior depends on information and processes extending beyond the model itself.

Component-level verification examines whether individual models, software functions, and communication mechanisms satisfy specified requirements, while validation examines whether they are suitable for their intended engineering applications. System-level verification examines whether integrated components and their interfaces satisfy the requirements governing their interaction. A fault-diagnosis model may perform reliably on representative measurements, for example, while its operational use depends on the correct association of observations with physical assets, the availability of relevant information, and the interpretation of diagnostic outputs by maintenance systems.

Digital twins illustrate the importance of continuing verification. A computational representation may be validated against experimental measurements or reference calculations, yet changes in equipment configuration can alter the relationship between the model and the physical asset. If these changes are not reflected in the digital representation, subsequent predictions may become unreliable even though the original validation remains valid for the earlier configuration.

Industrial AI faces related difficulties when models are transferred or updated. Changes in production conditions, sensor characteristics, and equipment behavior may alter input distributions and predictive uncertainty. Research on machine-learning safety has identified distribution shifts, robustness, and uncertainty among the unresolved problems affecting dependable model deployment [80]. Industrial verification must therefore examine the conditions under which a model remains applicable, rather than relying exclusively on its performance during development.

These considerations extend to decision execution. Advisory systems allow operators to interpret recommendations before implementation, whereas automated systems may influence equipment or production processes more directly. The latter require explicit procedures for authorization, constraint checking, failure

handling, and recovery. System-level verification must consequently address information consistency, model validity, operational feasibility, and the traceability of implemented actions.

Public datasets such as C-MAPSS, CWRU, MIMII, and MVTec AD support the evaluation of selected diagnostic and inspection tasks [61–64]. They do not reproduce the complete information flows, model updates, communication disturbances, and operational responsibilities of an integrated industrial intelligence system. Future evaluation frameworks should complement algorithm-oriented benchmarks with industrial scenarios involving inconsistent data, equipment changes, operational constraints, execution failures, and recovery procedures.

The objective is not to replace established predictive evaluation, but to connect it with evidence of reliable system operation. This distinction becomes increasingly important as intelligent functions move from monitoring and advisory applications toward decisions with direct consequences for industrial processes.

4.4 Security, Functional Safety, and Decision Accountability

The integration of industrial networks, engineering software, digital twins, and AI increases the exchange of information between OT and enterprise computing environments. Although this connectivity supports broader monitoring and coordination, it also creates pathways through which unauthorized access, manipulated information, or compromised software components may affect physical operations.

Industrial cybersecurity must therefore account for the relationship between information security and production continuity. IEC 62443-3-3 specifies system security requirements and security levels for industrial automation and control systems [36]. Functional safety addresses a different aspect of system assurance. IEC 61508-1 establishes general requirements for the functional safety of electrical, electronic, and programmable electronic safety-related systems [81]. These standards concern different sources of risk, and compliance with one does not establish satisfaction of the other.

AI introduces additional vulnerabilities associated with data manipulation, model behavior, and dependencies on external information or software tools. Research on adversarial machine learning has examined attacks affecting learning and inference, while MITRE ATLAS provides structured information concerning adversarial techniques directed at AI systems [82, 83]. In industrial environments, the consequences of these vulnerabilities depend on the physical processes and operational decisions influenced by the affected models.

Generative and agent-based systems require particular attention when their outputs include executable instructions, software configurations, or sequences of tool interactions. Research on language-model-based agents describes planning and external-tool interaction, which require additional safeguards in industrial use [84]. A generated recommendation may appear technically plausible without satisfying equipment restrictions or operational procedures. Similarly, individually permissible actions may produce unsuitable results when executed in combination. Candidate actions must therefore remain distinct from decisions authorized for industrial implementation.

Decision accountability becomes more difficult when recommendations depend on several interconnected components. A maintenance or control decision may draw on sensor measurements, an engineering model, a learned predictor, a digital-twin simulation, and operational constraints. Reconstructing the basis of that decision requires information about the inputs, model versions, assumptions, and authorization procedures involved. Traceability must therefore extend beyond recording the final output.

Security, functional safety, and decision accountability are related but distinct requirements. Cybersecurity concerns the integrity, availability, and authorized operation of connected systems. Functional safety concerns risks arising from failures of safety-related functions. Decision accountability concerns the traceability and justification of actions supported by intelligent systems. Their coordinated treatment is necessary when computational recommendations are incorporated into industrial operations.

4.5 Research Priorities for Integrated Industrial Intelligence

The preceding analysis identifies five research priorities arising from the interaction of industrial intelligence technologies. They concern the reliability of relationships among information systems, computational models, and operational decisions rather than improvements in isolated technological capabilities.

The first priority is semantic interoperability across industrial information systems. Although communication standards support information exchange, industrial intelligence requires measurements, equipment identities, configurations, and engineering records to retain consistent meanings across platforms and lifecycle stages. Research should therefore examine semantic representations and traceability mechanisms that remain reliable as equipment and data sources change.

The second priority is verifiable mechanism–data modeling. Existing approaches incorporate physical knowledge into learning in different ways, but their reliability depends on assumptions, uncertainty, and operating-domain coverage. Further work should examine how physical constraints, learned approximations, and uncertainty estimates can be combined while maintaining identifiable conditions of model validity, particularly when models are transferred between equipment populations or updated during operation.

The third priority concerns system-level evaluation. Algorithm-oriented benchmarks have supported methodological development, but they provide limited evidence concerning dependencies among data pipelines, engineering models, communication systems, and decision execution. Evaluation frameworks should therefore incorporate data inconsistencies, model updates, operational disturbances, decision constraints, and recovery procedures. Such assessments would help distinguish failures originating within predictive models from those introduced through system integration.

The fourth priority is the verification of intelligent decisions before execution. Industrial applications increasingly involve automated planning, optimization, and interactions between AI and engineering software. Research is needed on the representation of operational constraints, uncertainty-aware decision assessment, and procedures for determining whether proposed actions remain feasible under current production conditions.

The fifth priority concerns security and accountability across interconnected technological components. As industrial intelligence becomes more dependent on shared data and computational services, vulnerabilities may affect several parts of the operational architecture. Research should address model and data provenance, the verification of software-tool interactions, traceable authorization, and recovery when intelligent functions behave unexpectedly.

Figure 5 summarizes the relationships among the five deployment barriers, the corresponding research gaps, and the research directions identified in this review. They address complementary requirements of industrial intelligence and should not be interpreted as independent technological development paths. Semantic interoperability supports reliable information exchange, mechanism–data integration connects computational models with engineering knowledge, and system-level verification examines whether these capabilities remain dependable during operation. Decision verification and accountability extend these requirements to the actions supported by intelligent systems.

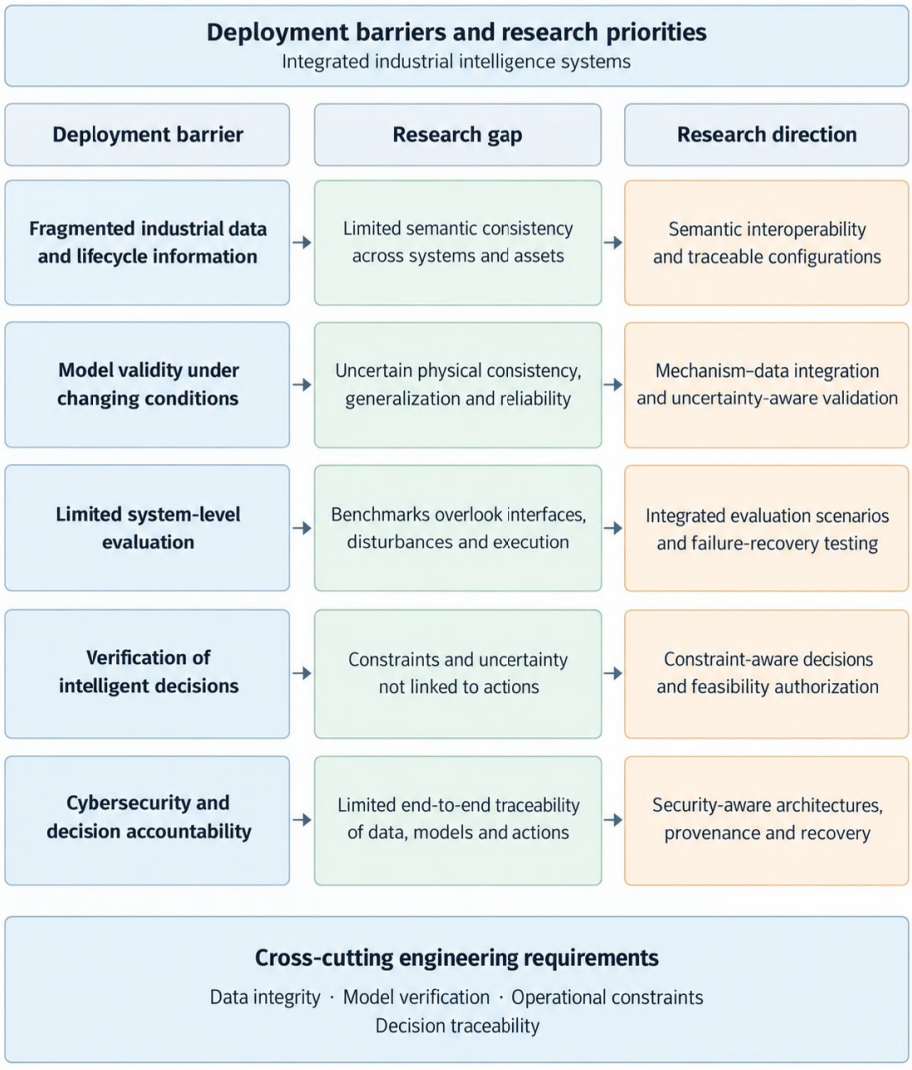


Figure 5. Deployment barriers, research gaps, and corresponding research priorities for integrated industrial intelligence systems

Table 5 provides the corresponding deployment barriers, unresolved research gaps, and proposed research directions. It retains the detailed engineering considerations underlying the five priorities, while Figure 5 provides their overall conceptual organization.

Table 5. Deployment barriers, unresolved research gaps, and proposed research directions for industrial intelligence systems

Deployment Barrier	Research Gap	Future Research Direction
Heterogeneous industrial data and fragmented lifecycle information	Insufficient semantic consistency across equipment, engineering models, and operational records	Digital-thread-based data governance, semantic interoperability, and traceable configuration management
Uncertain validity of computational models under changing conditions	Incomplete methods for assessing model reliability, physical consistency, and uncertainty beyond the conditions represented during development	Verifiable mechanism–data integration, uncertainty-aware modeling, and continuous assessment of model validity
Limited system-level evaluation of industrial intelligence	Existing benchmarks emphasize individual algorithms and provide insufficient coverage of cross-system dependencies, operational disturbances, and decision execution	System-level benchmarks incorporating data consistency, model updates, communication disturbances, operational constraints, and failure recovery
Difficulty verifying intelligent decisions before execution	Incomplete integration of physical constraints, operational requirements, and uncertainty assessment into automated decision processes	Constraint-aware decision-making, feasibility verification, and traceable authorization of intelligent actions
Cybersecurity vulnerabilities and limited accountability across interconnected systems	Insufficient methods for tracing vulnerabilities and reconstructing decisions across data pipelines, computational models, and operational interfaces	Security-aware industrial intelligence architectures, decision provenance, version traceability, and verifiable recovery mechanisms

The research priorities identified in Table 5 are interconnected. A physically consistent predictive model may still produce unsuitable recommendations when equipment configuration information is outdated, while reliable information exchange cannot compensate for a model operating beyond its validated conditions. Similarly, a feasible computational recommendation may not be appropriate for implementation without authorization and consideration of production constraints.

The next stage of industrial intelligence research should therefore place greater emphasis on the verifiability of integrated functions. Reliable operation requires evidence that information remains interpretable, models remain applicable, and intelligent decisions satisfy the constraints of the physical process. This system-oriented perspective connects technological capability with engineering trustworthiness and provides the basis for the conclusions in Section 5.

5 Conclusions

This study examined the technological foundations of industrial intelligence through a structured review of industrial software, the industrial Internet, digital twins and digital threads, and industrial AI. Using a seven-dimensional analytical framework, the review compared their engineering characteristics, operational requirements, and integration constraints. The analysis showed that industrial intelligence readiness cannot be determined solely by the maturity or performance of individual technologies. It also depends on the consistency of industrial information, the validity of engineering models, the availability of computational resources, and the reliability of decision execution.

The comparison identified two recurring categories of deployment constraints. Industrial software and industrial communication infrastructures were primarily associated with legacy-system compatibility, proprietary interfaces, and heterogeneous information structures. Digital twins and industrial AI introduced additional requirements concerning model fidelity, uncertainty, robustness, and the verification of intelligent decisions. These categories are not mutually exclusive, but they help distinguish the integration of established engineering capabilities from the verification of computational functions under changing operating conditions.

Four interdependent requirements emerged from the analysis: data consistency, model validity, computational adequacy, and decision-execution reliability. Limitations in one area can affect the performance of the integrated

system even when individual components satisfy their own evaluation criteria. Mechanism–data integration provides a means of connecting engineering knowledge with computational learning, but the inclusion of physical constraints does not independently establish model trustworthiness. Verification must also address operating-domain coverage, uncertainty, and the consequences of model outputs when they are used to support industrial actions.

The principal contribution of this review is a system-oriented framework that connects technology-specific capabilities with the engineering requirements of integrated industrial intelligence. It provides a basis for examining technological maturity alongside system dependencies and trustworthiness, while identifying semantic interoperability, verifiable mechanism–data modeling, system-level evaluation, constraint-aware decision-making, and decision traceability as related research priorities. These considerations are particularly relevant as industrial AI moves from monitoring and advisory functions toward decisions that influence physical production processes.

The findings are based on a qualitative synthesis of the selected literature rather than experimental validation of an integrated industrial intelligence system. Differences in source coverage, industrial applications, and available verification evidence limit direct comparisons across technological domains. The framework is therefore intended to support the interpretation of engineering requirements, not to provide a universal maturity ranking. Within these boundaries, the review establishes a basis for assessing how technological capabilities and system dependencies shape the reliable deployment of industrial intelligence.

Author Contributions

Conceptualization, L.Z. and B.L.; methodology, L.Z. and B.L.; validation, L.Z. and B.L.; formal analysis, L.Z.; investigation, L.Z. and B.L.; resources, B.L.; data curation, L.Z.; writing—original draft preparation, L.Z.; writing—review and editing, B.L.; visualization, L.Z. and B.L.; supervision, B.L.; project administration, B.L. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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